Time Series Regression

David Puelz
The University of Austin

What are time series?

- Data where the cases represent time: data collected every day, month, year, etc.
- Time series are important for both explaining how variables change over time and forecasting the future
- Examples of time series data:
  - Google's closing daily stock price every day in 2020
  - Inventory levels of each item at a retail store at the end of every week in 2020
  - Number of new COVID cases in the US each day since the start of the pandemic
  - Apple's quarterly revenue since 2009

Anatomy of a time series

Some notation:

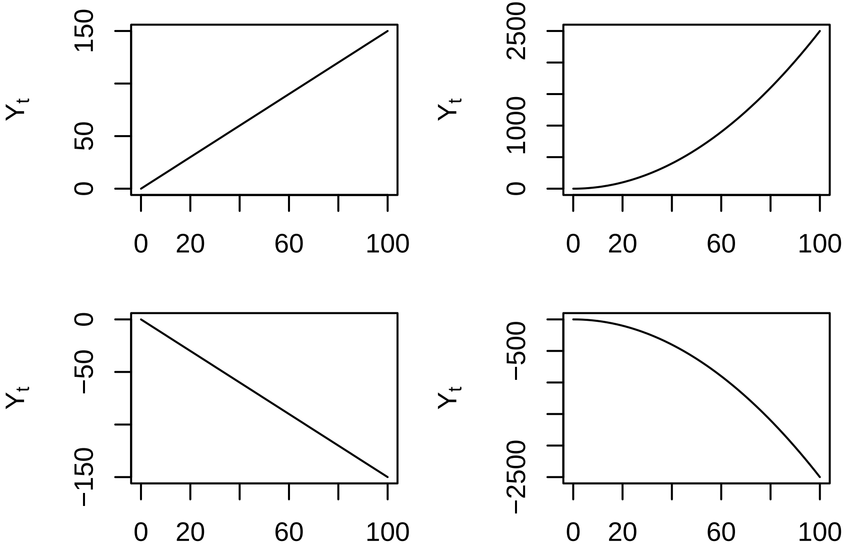
$$t = \text{time}$$
$$Y_t = \text{the value of the variable we are interested in, at time } t$$

Y_t may be composed of one or more components:

- Trend
- Seasonal
- Cyclical
- Random

Trend component

An trend is persistent upwards or downwards movement in the data

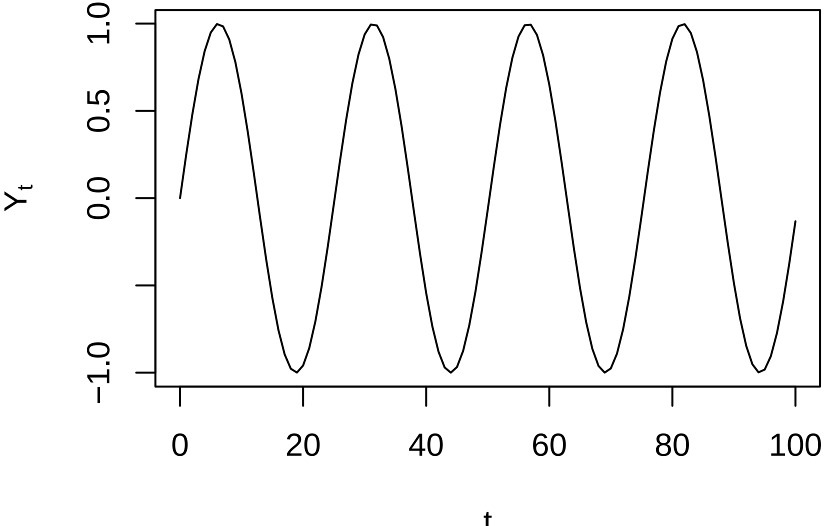


Trend component

- The trend doesn't need to be linear!
- Example: Moore's Law (accelerating increase of transistor count)
- Example: US population
- A time series with no trend is stationary

Seasonal component

Seasonal fluctuation occurs when predictable up or down movements occur *over a regular interval*.

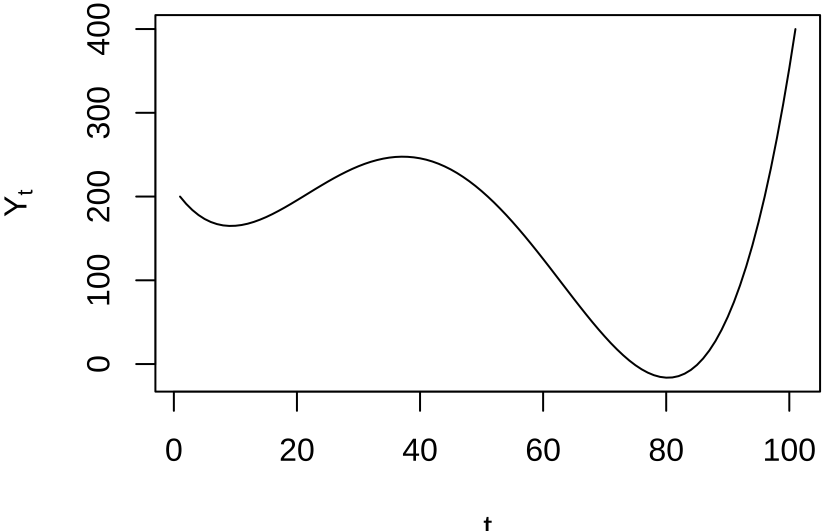


Seasonal component

- The ups and downs must occur over a regular interval (e.g., every month, or every year)
- Example: Highway traffic volume is highest during rush hour every day
- Example: Supermarket sales may be highest every month right after common paydays like the 15th and 30th

Cyclic component

Cyclic fluctuations occur at unpredictable intervals, e.g. due to changing business or economic conditions.

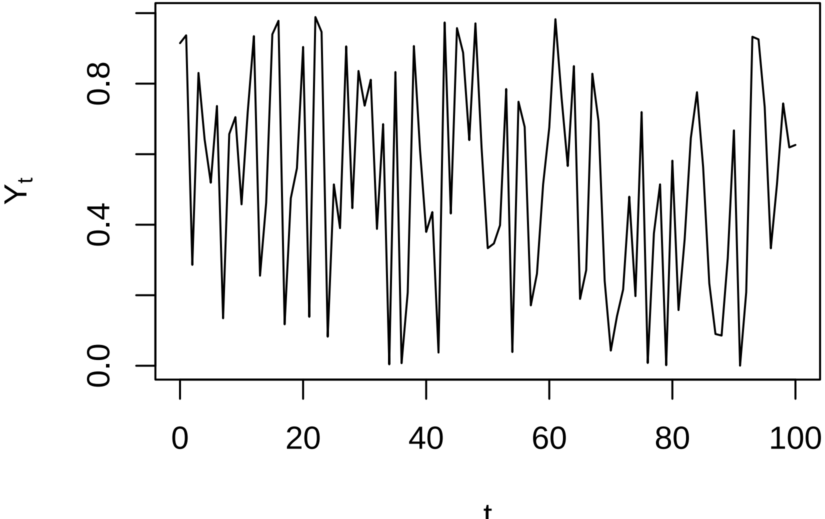


Cyclic component

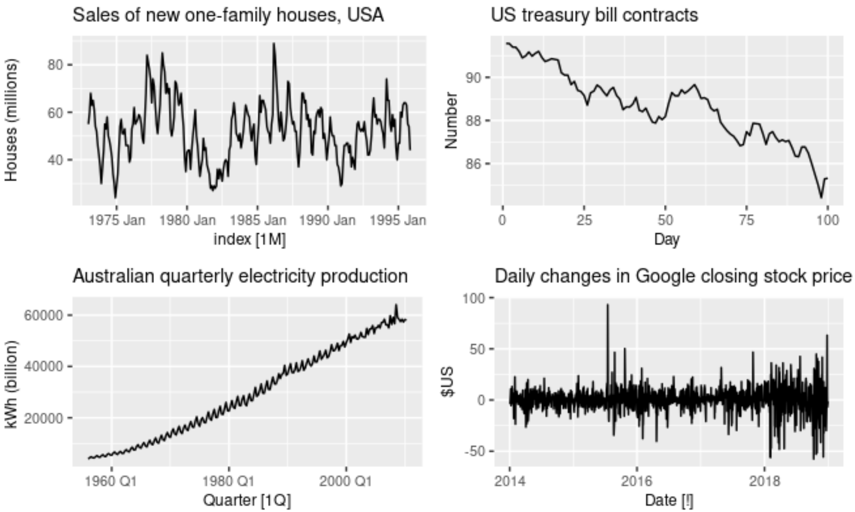
- In contrast to seasonal fluctuations, cyclic fluctuations do not occur at regular, predictable intervals
- It may be possible to predict cyclic components based on some other (non-time) variable
- Example: Restaurant sales dropped dramatically in 2020 due to COVID, as people ate out less
- Example: Sales of bell bottoms rose in the 60s and 70s, declined by the 80s, and then had a resurgence in the 90s

Remainder/Error component

Any real time series will always have random noise as well, which can't be predicted or forecast.



Which component(s) do you see?



Putting these together

Real time series will usually include a combination of these four components. We will model the time series Y_t either additively:

$$Y_t = \text{trend} + \text{seasonal} + \text{random} = T_t + S_t + E_t$$

Or multiplicatively:

$$Y_t = \text{trend} \cdot \text{seasonal} \cdot \text{random} = T_t \cdot S_t \cdot E_t$$

(E_t consists of both the cyclic and error components, as both are unpredictable.)

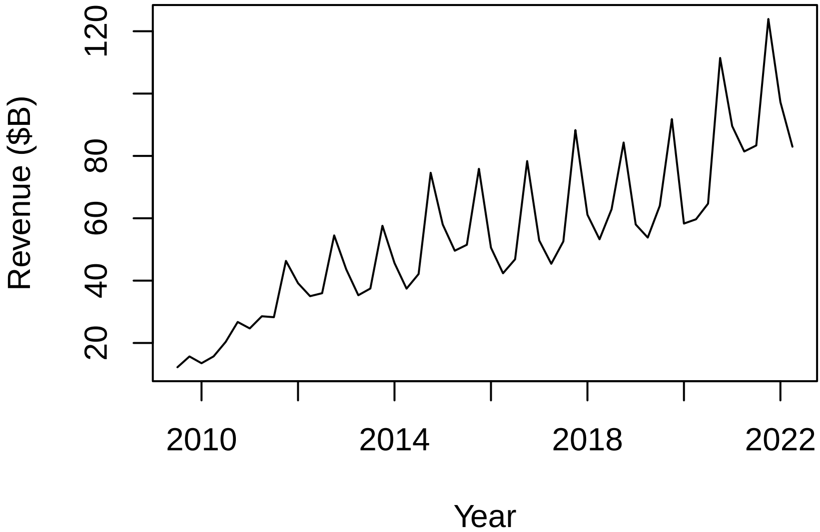
Additive models

- All components (T_t, S_t, E_t) are in the units of Y
- Most appropriate when seasonal fluctuations are consistent (do not increase or decrease over time)
- The trend component T_t is a function of t (e.g., linear or quadratic)
- The seasonal component S_t is a set of dummy variables representing “seasons”
- So we can estimate additive models using regular old regression!

Additive decomposition

1. Run a regression predicting Y as a function of:
   - t, t^2, $\log(t)$ etc (the trend component T_t)
   - Dummy variables for the seasons (the seasonal component S_t)
2. To make a prediction for Y, plug into the model!
3. The residuals of this model correspond to the error component E_t

Apple quarterly revenue



What components do you see here?

Apple data

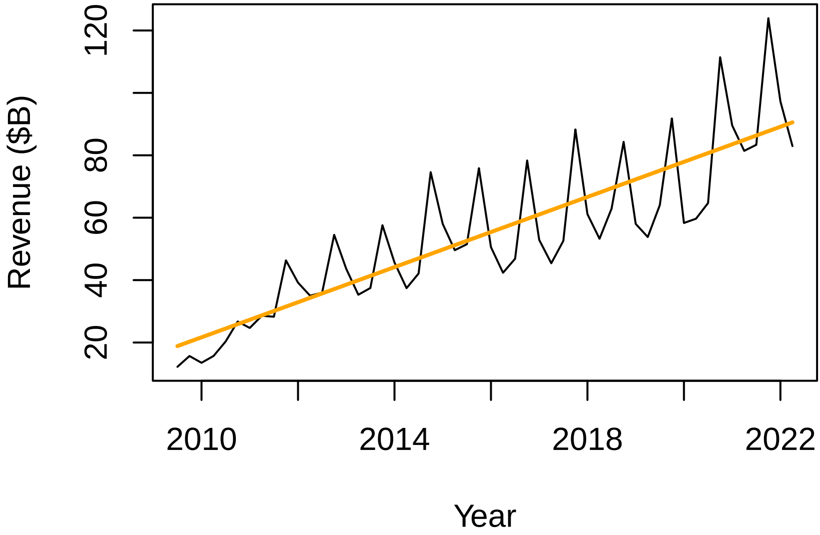
```r
head(apple)
```

```
  Period    Time Year Quarter Revenue
1      1 2009.50 2009      Q3   12.21
2      2 2009.75 2009      Q4   15.68
3      3 2010.00 2010      Q1   13.50
4      4 2010.25 2010      Q2   15.70
5      5 2010.50 2010      Q3   20.34
6      6 2010.75 2010      Q4   26.74
```

(Note that the fractional part of year indicates the quarter.)

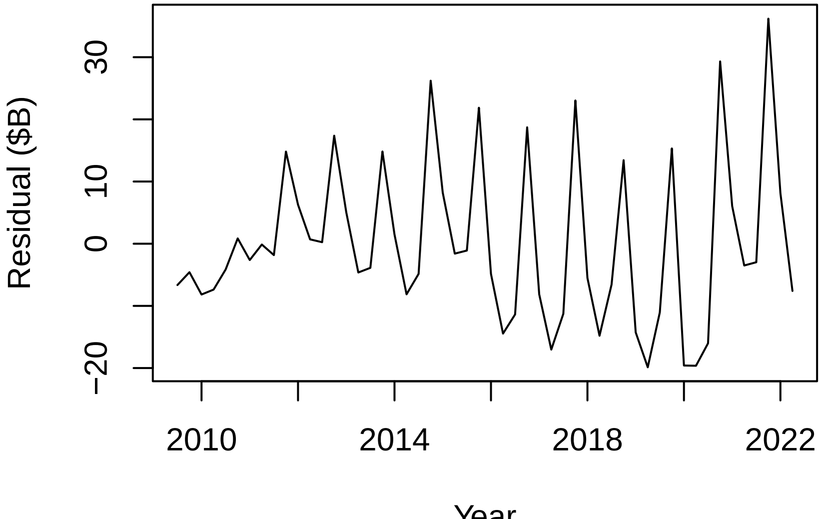
```
linear.trend <- lm(Revenue ~ Period, data=apple)
summary(linear.trend)


Call:
lm(formula = Revenue ~ Period, data = apple)

Residuals:
   Min     1Q Median     3Q    Max
-19.88  -8.13  -3.24   6.74  36.20

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   17.452      3.758    4.64  2.5e-05 ***
Period         1.406      0.123   11.39  1.7e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.4 on 50 degrees of freedom
Multiple R-squared:  0.722,Adjusted R-squared:  0.716
F-statistic:  130 on 1 and 50 DF,  p-value: 1.67e-15
```

Fitting a linear trend component



Detrended data

The residuals from this model show the "detrended" data (but there's still some trend left, and clear seasonality that we are ignoring!):

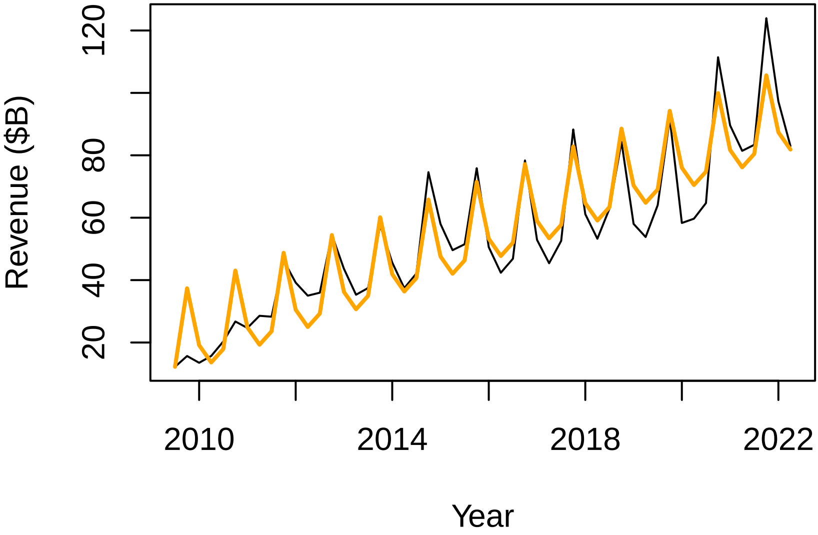


Adding seasonality

Let's incorporate seasonality into our model, by adding dummies for the quarter (leaving out one, like any categorical variable):

```
linear.trend.seasonality <- lm(Revenue ~ Period + Quarter, data=apple)
summary(linear.trend.seasonality)


Call:
lm(formula = Revenue ~ Period + Quarter, data = apple)

Residuals:
   Min     1Q Median     3Q    Max
-21.70  -5.13   1.09   5.30  18.33

Coefficients:
            Estimate Std. Error t value   Pr(>|t|)
(Intercept)  14.8822     3.1273    4.76 0.00001892 ***
Period        1.4215     0.0776   18.32    < 2e-16 ***
QuarterQ2    -6.9492     3.2850   -2.12       0.04 *
QuarterQ3    -4.0878     3.2878   -1.24       0.22
QuarterQ4    19.6569     3.2850    5.98 0.00000028 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.37 on 47 degrees of freedom
```

Linear trend with seasonality



Interpreting the coefficients

```
             Estimate Std. Error t value   Pr(>|t|)
(Intercept)  14.8822     3.1273    4.76 0.00001892 ***
Period        1.4215     0.0776   18.32    < 2e-16 ***
QuarterQ2    -6.9492     3.2850   -2.12       0.04 *
QuarterQ3    -4.0878     3.2878   -1.24       0.22
QuarterQ4    19.6569     3.2850    5.98 0.00000028 ***
```

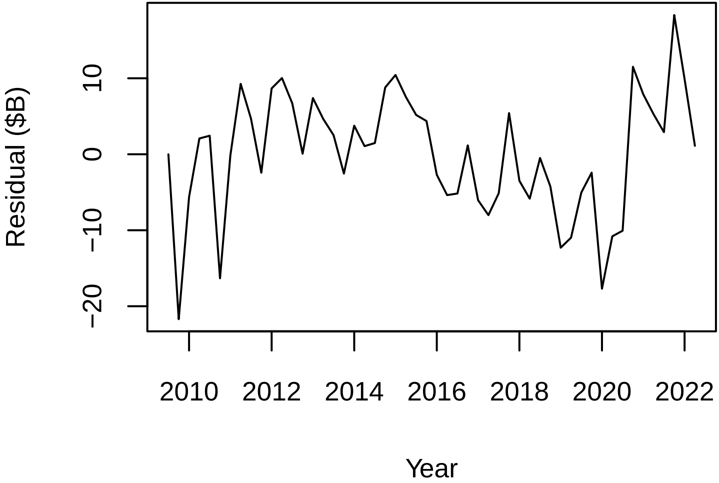
- Q2's are expected to be \$6.9B worse than Q1's
- Q3's are expected to be \$4.1B worse than Q1's
- Q4's are expected to be \$19.7B better than Q1's

What does our model predict for the future?

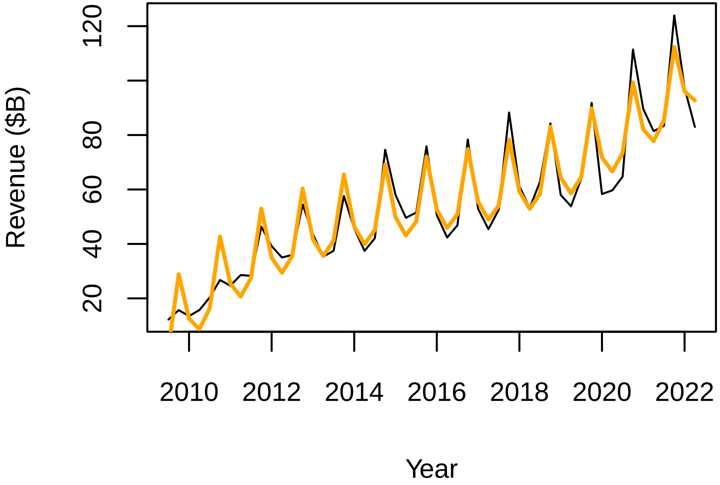
- What does the final model predict for Apple in 2023 Q1?
- What does the final model predict for Apple in 2030 Q1? (Should we trust that prediction?)

Detrended and seasonalized data

The residuals from this model show the “detrended and deasonalized” data (but there’s still some trend left!):

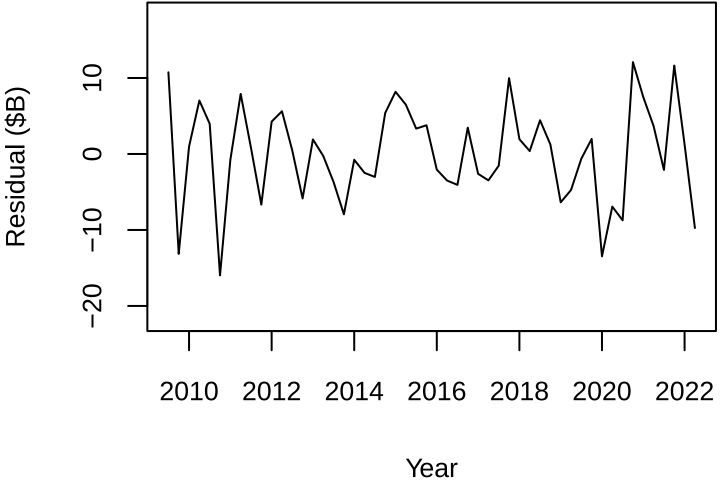


Improving the model by fitting a cubic trend



Detrended and deasonalized data (cubic model)

The residuals from this model show the "detrended and deasonalized" data—much better! What's left looks random:



Multiplicative models

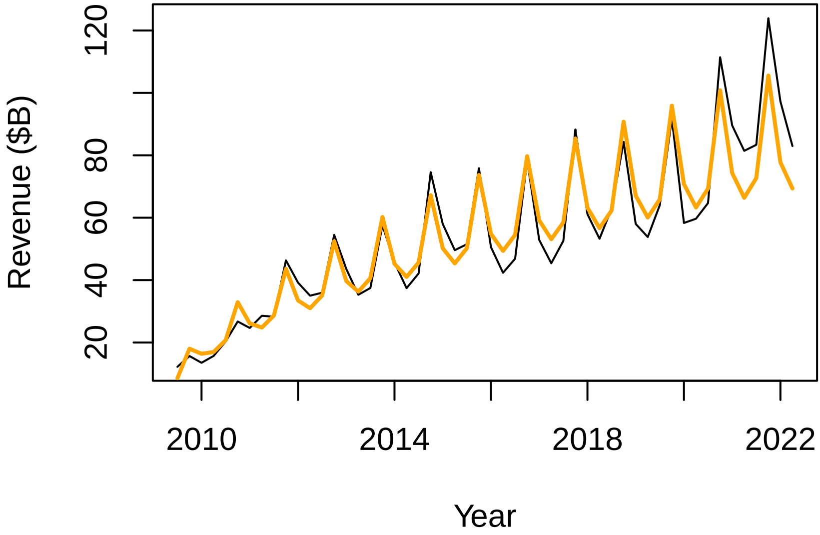
- Only the trend component T_t is in the units of Y
- The seasonal and error components are multipliers (e.g., applying a percentage rather than absolute change to the reference season)
- We can estimate this using regression by taking logs of both sides of the equation $Y_t = T_t \cdot S_t \cdot E_t$:

$$\log(Y_t) = \log(T_t) + \log(S_t) + \log(E_t)$$

- A multiplicative model is appropriate when the magnitude of the seasonality increases (or decreases) over time

```
log.trend.seasonality <- lm(log(Revenue) ~ log(Period) + Quarter, data=apple)
summary(log.trend.seasonality)


Call:
lm(formula = log(Revenue) ~ log(Period) + Quarter, data = apple)

Residuals:
    Min      1Q  Median      3Q     Max
-0.2083 -0.0810 -0.0236  0.0954  0.3540

Coefficients:
            Estimate Std. Error t value   Pr(>|t|)
(Intercept)   2.1936     0.0719   30.53    < 2e-16 ***
log(Period)   0.5493     0.0204   26.95    < 2e-16 ***
QuarterQ2    -0.1248     0.0507   -2.46      0.017 *
QuarterQ3    -0.0453     0.0508   -0.89      0.377
QuarterQ4     0.3166     0.0507    6.25 0.00000011 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.129 on 47 degrees of freedom
Multiple R-squared:  0.945,Adjusted R-squared:  0.94
F-statistic:  202 on 4 and 47 DF,  p-value: <2e-16
```

Multiplicative model captures increasing variation



Writing out a multiplicative model

Our model of the trend component is:

$$\log(\widehat{\mathsf{Rev}}) = 2.19 + 0.55 \log(t) - 0.12(\mathsf{Q2}) - 0.05(\mathsf{Q3}) + 0.32(\mathsf{Q4})$$

Interpreting multiplicative model coefficients

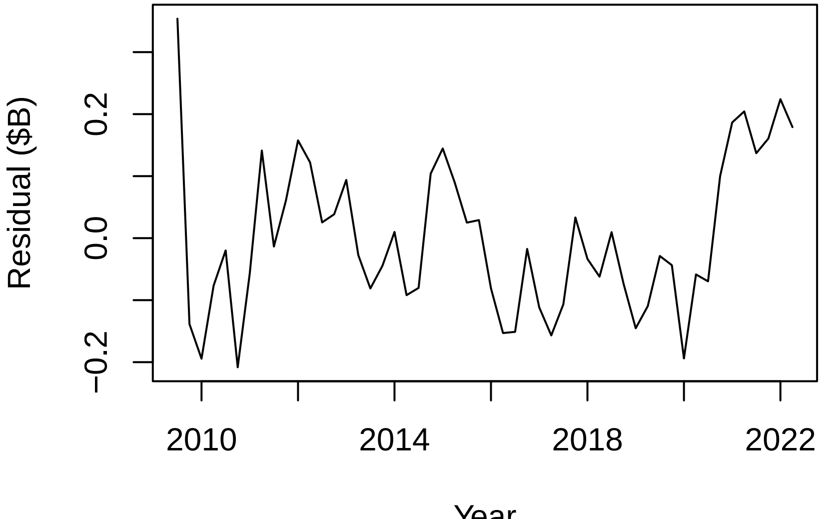
```
             Estimate Std. Error t value   Pr(>|t|)
(Intercept)   2.1936     0.0719   30.53    < 2e-16 ***
log(Period)   0.5493     0.0204   26.95    < 2e-16 ***
QuarterQ2    -0.1248     0.0507   -2.46      0.017 *
QuarterQ3    -0.0453     0.0508   -0.89      0.377
QuarterQ4     0.3166     0.0507    6.25 0.00000011 ***
```

Because this is a model where we are predicting log(Y):

- Q2's are expected to be 12% below Q1's
- Q3's are expected to be 5% below Q1's
- Q4's are expected to be 32% above Q1's

Detrended/deseasonalized (multiplicative)

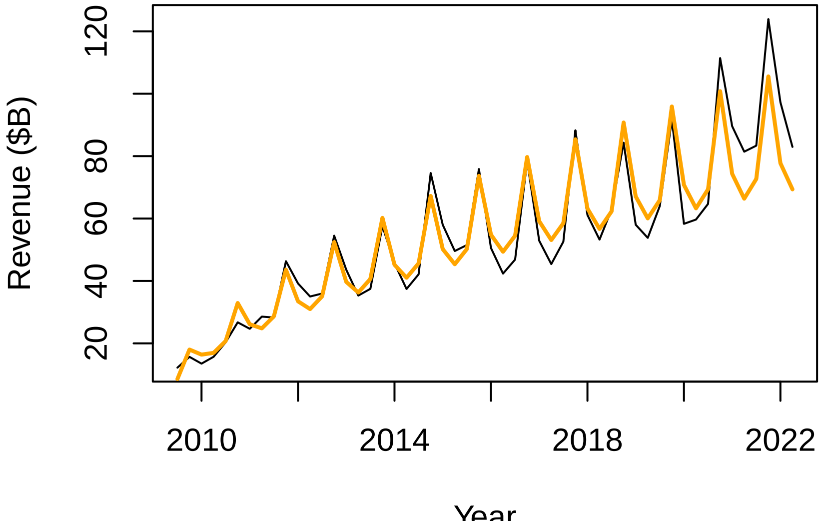
The residuals from this model show the "detrended and deasonalized" data—good! What's left looks (pretty) random:



Sometimes forecasting using a trend-based model is dangerous—things change!

Have things changed?

Apple seems to be consistently outperforming the model in the last couple of years.



Limitations of decomposition models

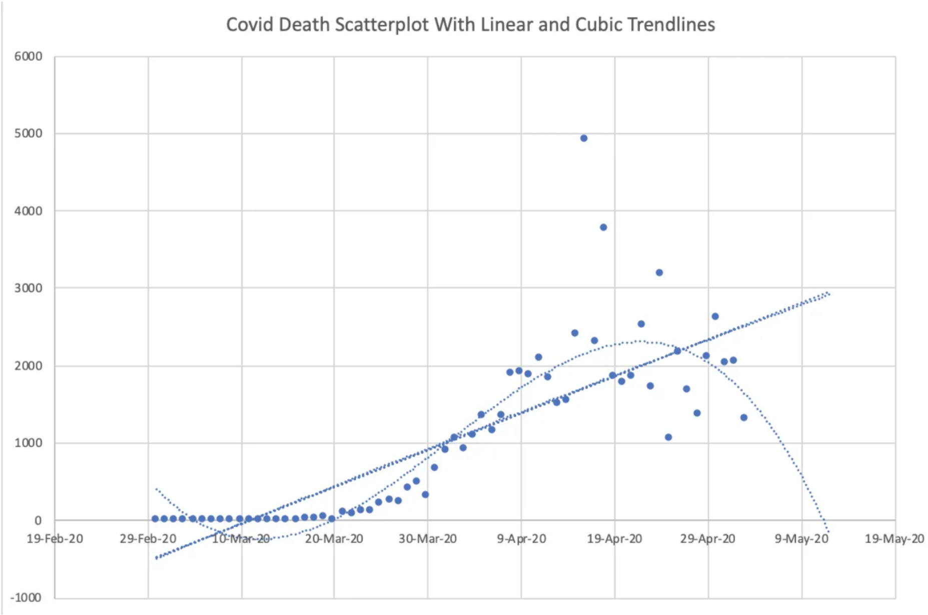
Previously, we created decomposition models that predict Y_t as a function of time:

$$Y_t = \text{trend} + \text{seasonal} + \text{error} = T_t + S_t + E_t$$

$$Y_t = \text{trend} \times \text{seasonal} \times \text{error} = T_t \times S_t \times E_t$$

But sometimes forecasting using a trend-based model is dangerous or unworkable!

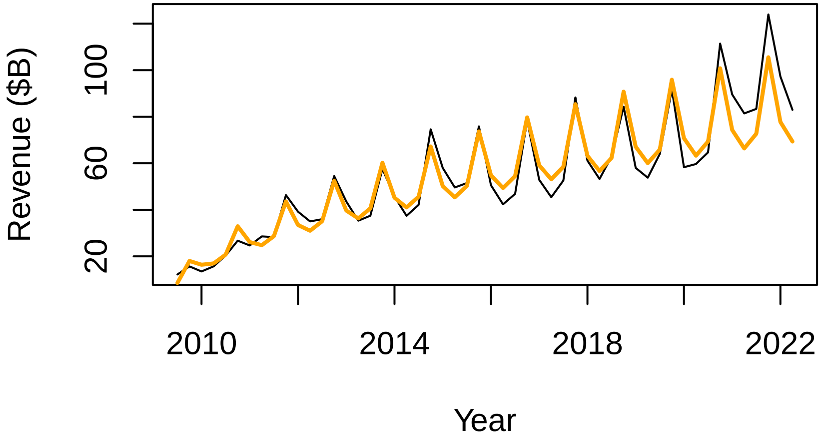
We might also be leaving some predictive ability on the table if E_t includes a cyclical component.



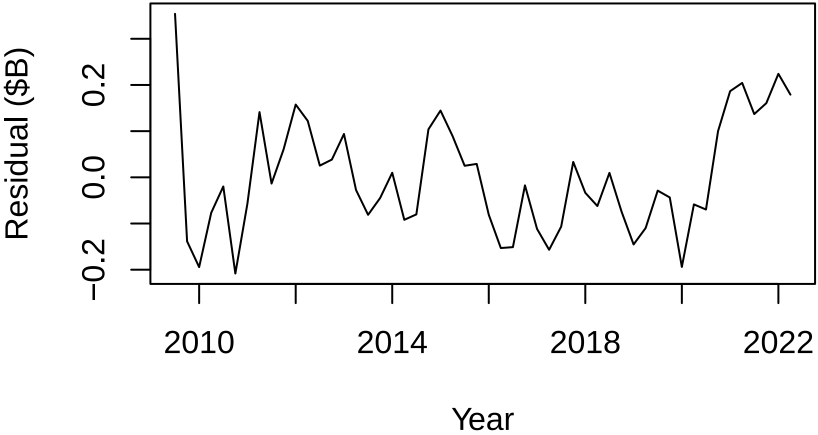
Covid Death Scatterplot With Linear and Cubic Trendlines
6000
5000
4000
3000
2000
1000
0
-1000
19-Feb-20
29-Feb-20
10-Mar-20
20-Mar-20
30-Mar-20
9-Apr-20
19-Apr-20
29-Apr-20
9-May-20
19-May-20

Recall: Multiplicative model for Apple's revenue

```
lm(log(Revenue) ~ log(Period) + Quarter, data=apple)
```

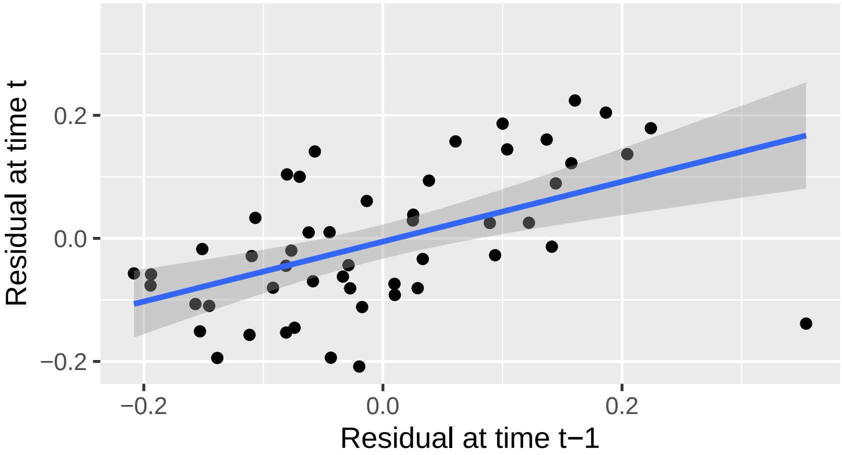


Recall: Multiplicative model for Apple's revenue



We thought our residuals looked pretty good, but...

Past prediction errors predict future errors

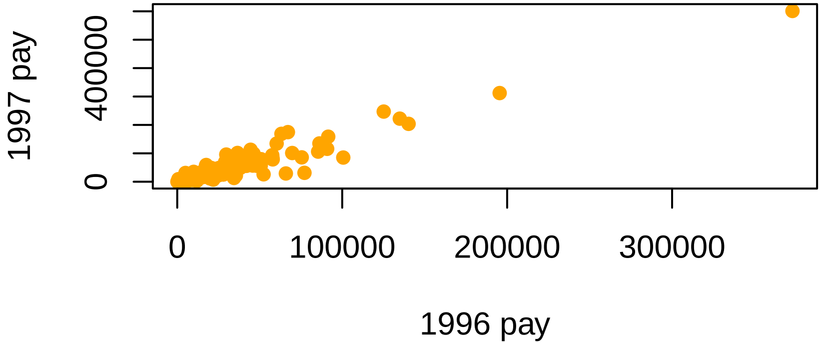


How can we use this to improve our model?

What factors do you think would be the best predictors of how much a company pays their CEO in a given year?

What factors do you think would be the best predictors of how much a company pays their CEO in a given year?

If we want to predict CEO pay in 1997, knowing the CEO's pay in 1996 seems to help a lot:



A model like this incorporates a lot of other variables that might not be available in our data set!

Autoregressive models

Key idea: Predict Y_t as a function of Y_{t-1}:

$$\hat{Y}_t = \hat{\beta}_0 + \hat{\beta}_1 Y_{t-1}$$

- Y_{t-1} is called the "1st lag" of Y (e.g., if $t = 1997$, then $Y_t =$ `pay1997` and $Y_{t-1} =$ `pay1996`)

Autoregressive models

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$$\hat{Y}_t = \hat{\beta}_0 + \hat{\beta}_1 Y_{t-1}$$

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- This is called autoregressive because it predicts the values of a time series based on previous values

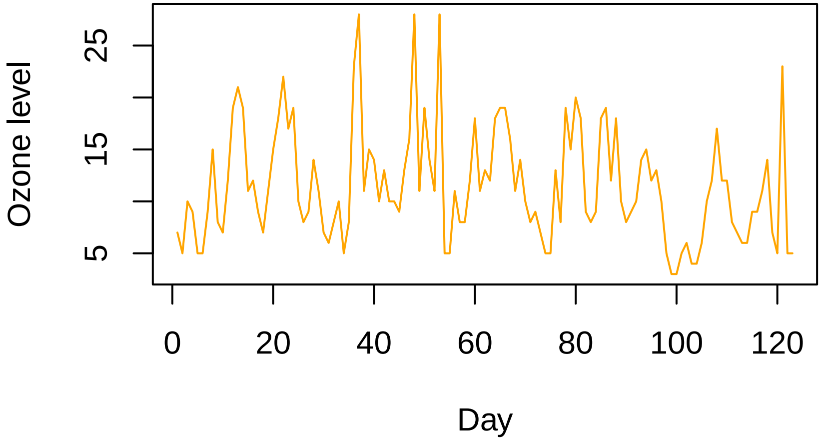
Autoregressive models

Key idea: Predict Y_t as a function of Y_{t-1}:

$$\hat{Y}_t = \hat{\beta}_0 + \hat{\beta}_1 Y_{t-1}$$

- Y_{t-1} is called the "1st lag" of Y (e.g., if $t = 1997$, then $Y_t =$ `pay1997` and $Y_{t-1} =$ `pay1996`)
- This is called autoregressive because it predicts the values of a time series based on previous values
- This is an AR(1) model (an AR(k) model would predict 1997 pay from the previous k years)

AR(1) example: Daily ozone in Houston

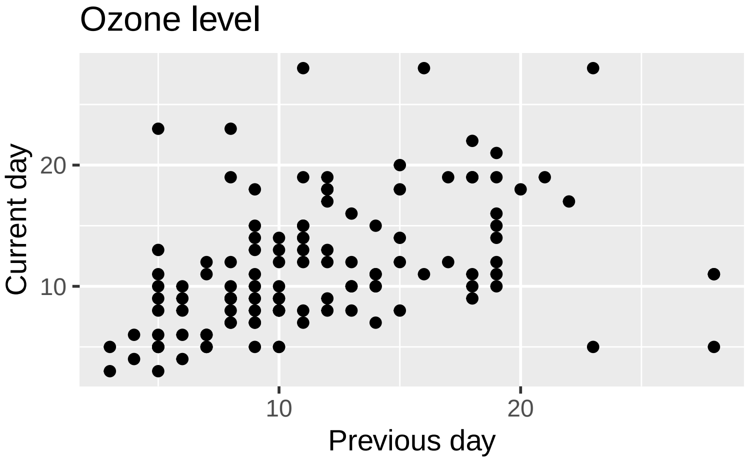


Decomposing Y_t into trend and seasonal components probably isn't going to work—no clear trend or seasonality!

```r
summary(lm(ozone ~ day, data=ozone))
```

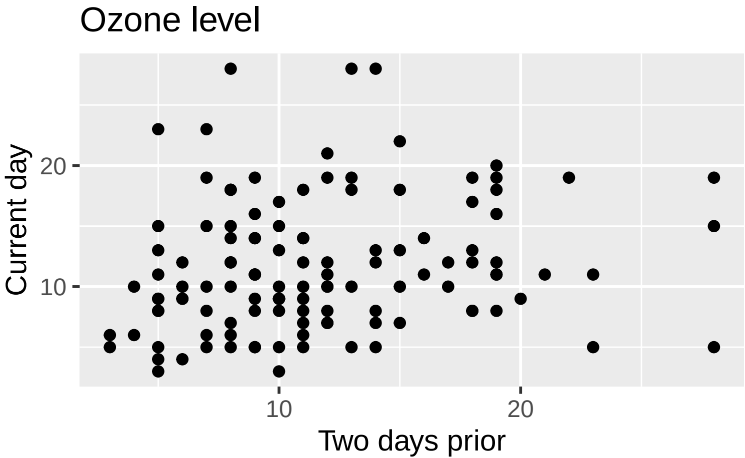
```

Call:
lm(formula = ozone ~ day, data = ozone)

Residuals:
   Min     1Q Median     3Q    Max 
 -7.97  -4.04  -1.09   2.79  16.27 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  13.0152     0.9772   13.32   <2e-16 ***
day          -0.0242     0.0137   -1.77    0.079 .  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.4 on 121 degrees of freedom
Multiple R-squared:  0.0253,Adjusted R-squared:  0.0173 
F-statistic: 3.14 on 1 and 121 DF,  p-value: 0.0788
```

Can we use yesterday's ozone level to predict today's?



$r = 0.4$

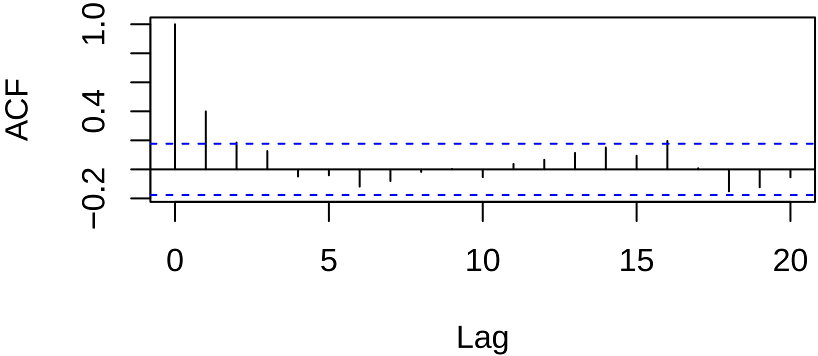
How about ozone from two days ago?



$r = 0.19$ (But can you think of any reason this correlation might be misleading?)

Autocorrelation is the correlation of Y_t with its lags (Y_{t-1}, Y_{t-2}, etc). The autocorrelation function (ACF) displays these:

```
acf(ozone$ozone)
```



Autocorrelations outside of the dashed blue lines are statistically significant.

```r
ozone <- ozone %>% mutate(lag1=lag(ozone), lag2=lag(ozone, 2))
ozone.model <- lm(ozone ~ lag1, data=ozone)
summary(ozone.model)
```

```
Call:
lm(formula = ozone ~ lag1, data = ozone)

Residuals:
   Min     1Q Median     3Q    Max
-13.19  -3.46  -1.11   2.68  16.68

Coefficients:
            Estimate Std. Error t value     Pr(>|t|)
(Intercept)   6.8745     1.0698    6.43 0.0000000028 ***
lag1          0.4042     0.0838    4.82 0.0000041974 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5 on 120 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.162,Adjusted R-squared:  0.155
F-statistic: 23.3 on 1 and 120 DF,  p-value: 0.0000042
```

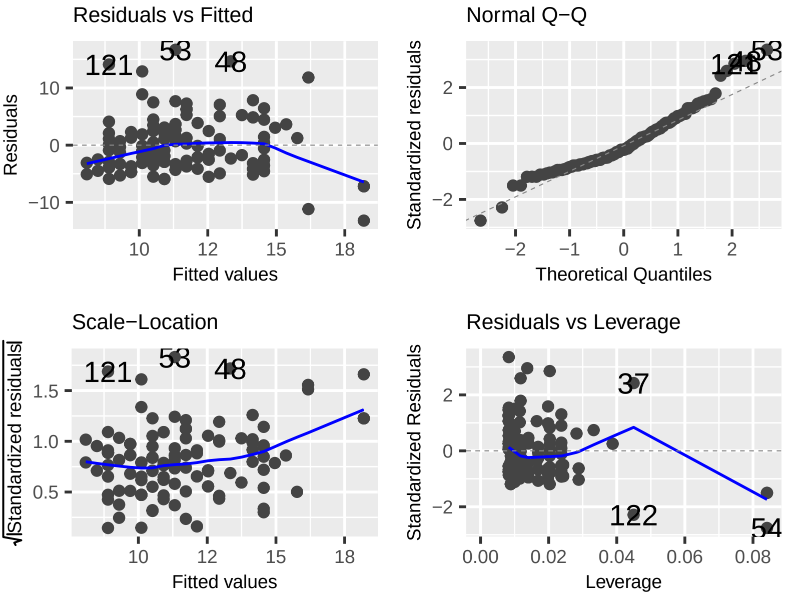
```
ozone <- ozone %>% mutate(lag1=lag(ozone), lag2=lag(ozone, 2))
ozone.model2 <- lm(ozone ~ lag1 + lag2, data=ozone)
summary(ozone.model2)
```

```

Call:
lm(formula = ozone ~ lag1 + lag2, data = ozone)

Residuals:
   Min     1Q Median     3Q    Max
-12.87  -3.35  -1.12   2.73  16.54

Coefficients:
            Estimate Std. Error t value   Pr(>|t|)
(Intercept)   6.7241     1.2604    5.33 0.00000047 ***
lag1          0.3835     0.0919    4.17 0.00005763 ***
lag2          0.0370     0.0922    0.40       0.69
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5 on 118 degrees of freedom
  (2 observations deleted due to missingness)
Multiple R-squared:  0.16,Adjusted R-squared:  0.145
F-statistic: 11.2 on 2 and 118 DF,  p-value: 0.000035
```

Checking assumptions of an AR(1) model

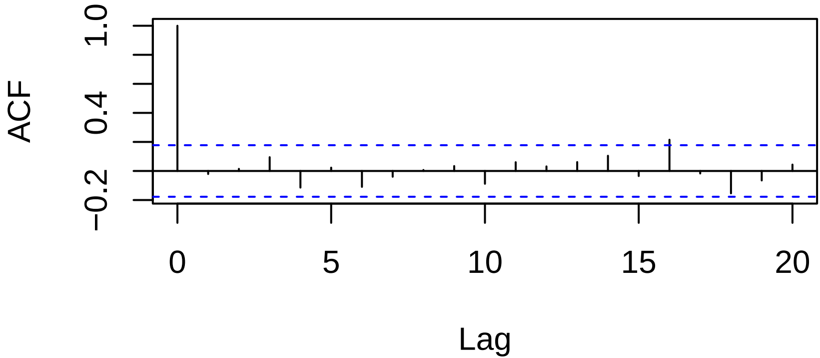
- Linearity, Homoscedasticity, Normality: Check using residual plot (linearity + homoscedasticity), Q-Q plot (normality), scale/location (homoscedasticity) like any other regression model
- Independence: Since this is a time series, we can actually check this by looking at the autocorrelation of the *residuals* (we want *no* significant autocorrelation)

Checking assumptions of an AR(1) model



Checking independence in an AR(1) model

```r
acf(residuals(ozone.model))
```



We expect 5% of autocorrelations to be significant just by chance at $\alpha = 0.05$, so this ACF suggests we are OK on independence!

Special kinds of AR(1) models

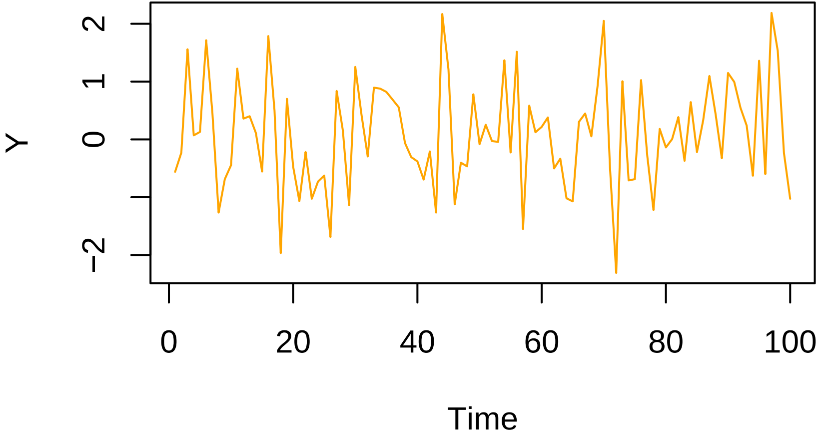
$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \epsilon_t$$

- If $\beta_0 = 0$ and $\beta_1 = 0$, then this is white noise
- If $\beta_0 \neq 0$ and $\beta_1 = 0$, then this is a random sample
- If $\beta_0 = 0$ and $\beta_1 = 1$, then this is a random walk
- If $\beta_0 \neq 0$ and $\beta_1 = 1$, then this is a random walk with drift
- If $\beta_1 < 1$, then Y_t will oscillate around the mean

White noise

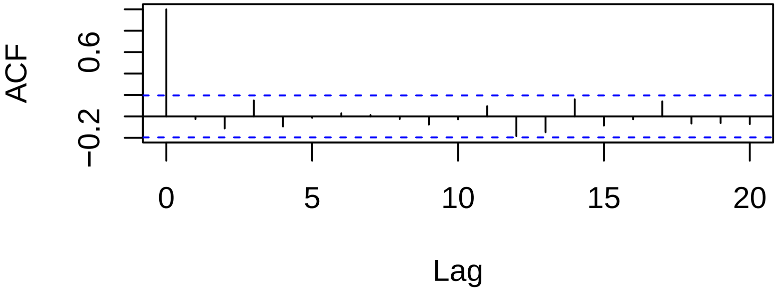
A white noise time series is one where there is no pattern at all—each value is totally unpredictable.

$$Y_t = \epsilon_t$$



Could my time series be white noise? If:

1. The ACF shows no significant autocorrelation:



2. And we fit an AR(1) model and find that that 0 falls within the 95% CIs for both β_0 and β_1:

```
              2.5 % 97.5 %
(Intercept) -0.0849  0.284
lag.y       -0.2286  0.176
```

Then there is no statistically significant evidence that the series isn't white noise

Random sample

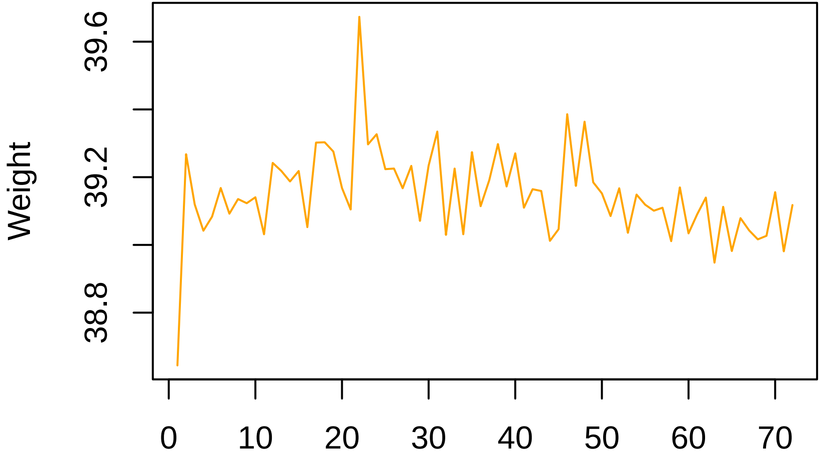
A random sample time series is one where each value is just independently sampled from the same distribution:

$$Y_t = \beta_0 + \epsilon_t$$

- There is no Y_{t-1} term: like with white noise, it doesn't matter what the previous value was
- β_0 represents the mean of the distribution that the values are being sampled from

Random sample

In the Folgers factory, 39-oz coffee cans are filled up with slightly different amounts of coffee each hour (just due to random variation with the machine):



Could it be a random sample?

1. The ACF shows no significant autocorrelation
2. We fit an AR(1) model and find that 0 falls within the 95% CI for β_1 but *not* for β_0:

```
              2.5 % 97.5 %
(Intercept) 27.482 44.112
lag.wt      -0.127  0.298
```

Then:

- There is statistically significant evidence that the series isn't white noise
- There is no statistically significant evidence that the series isn't a random sample

Random walk

A random walk time series is one where each value is predicted to be the previous value, plus some drift (maybe):

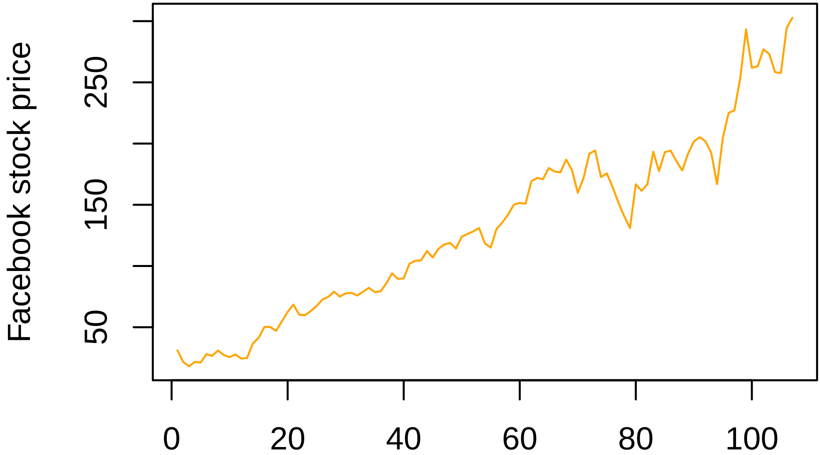
$$Y_t = \beta_0 + Y_{t-1} + \epsilon_t$$

- β_0 is the "drift" (could be 0) and is in some sense the overall trend
- A time series is a random walk if the *changes* are a random sample time series:

$$Y_t - Y_{t-1} = \beta_0 + \epsilon_t$$

Facebook stock price

Random walks are often a good choice for stock prices, like the Facebook stock price over a 107-month period from June 2012 to last year:



Could it be a random walk?

We fit an AR(1) model and find that 1 falls within the 95% CI for β_1 (and zero doesn't):

```
              2.5 % 97.5 %
(Intercept) -2.766   6.64
lagFB        0.973   1.04
```

Then:

- There is statistically significant evidence that the series isn't white noise or a random sample
- There is no statistically significant evidence that the series isn't a random walk

Bonus: If this series isn't truly a random walk, do you think it might behave similarly to a random walk anyway? Why?

Basic time series concepts

Additive decomposition models

Multiplicative decomposition models

A warning about trend-based forecasting

Autoregressive models

Special kinds of AR(1) models

Making predictions with AR models

Combining decomposition and AR models

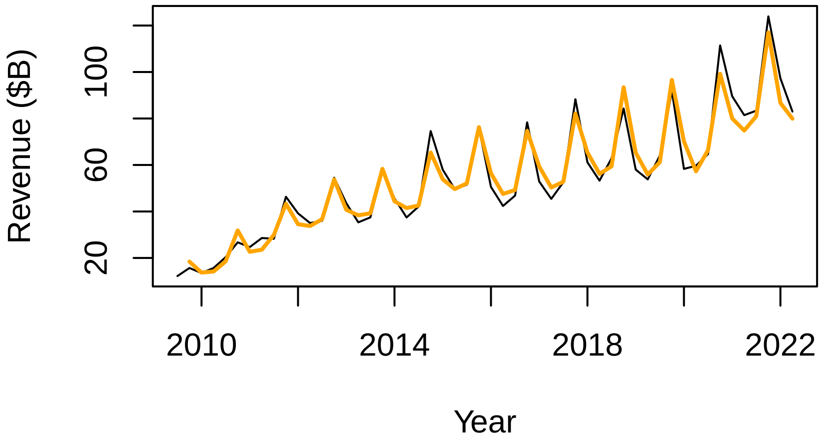
The market model

Making predictions in time series

Type	Model	Predicted Y_t
White noise	$Y_t = \epsilon_t$	0
Random sample	$Y_t = \beta_0 + \epsilon_t$	$\hat{\beta}_0$ (or average Y)
Random walk	$Y_t = \beta_0 + Y_{t-1} + \epsilon_t$	$\hat{\beta}_0 + Y_{t-1}$
General AR(1)	$Y_t = \beta_0 + \beta_1 Y_{t-1} + \epsilon_t$	$\hat{\beta}_0 + \hat{\beta}_1 Y_{t-1}$

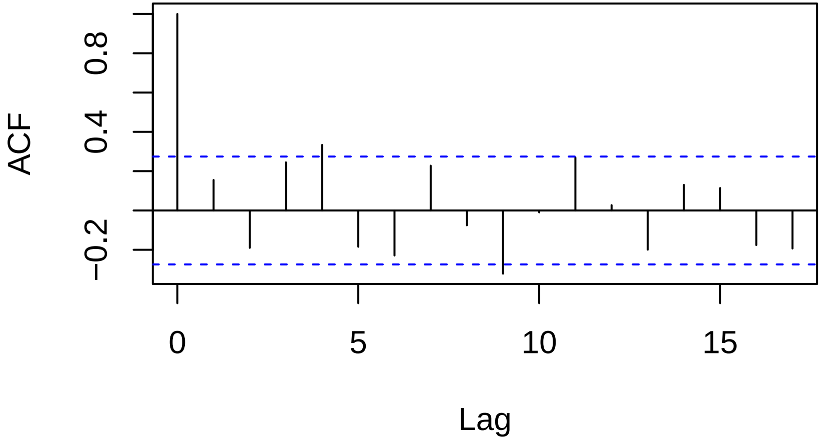
Combining decomposition and AR models

```
apple = apple %>% mutate(lag1 = lag(Revenue))
apple_model <- lm(log(Revenue) ~ log(Period) + Quarter
                  + log(lag1), data=apple)
```



One lag fixes residual autocorrelation

```
acf(residuals(apple_model))
```



Why lags and not residuals?

- We started with the observation that the residuals (prediction errors) were autocorrelated. So why not use lagged *residuals* as predictors?
- We need a model to compute residuals, so the model can't depend on the residuals!
- Y_{t-1} (in this case $\log(Y_{t-1})$) contains all the relevant predictive information to eliminate lag-1 autocorrelation in the residuals

Summary: Building a time series model

- Start with obvious seasonal/trend components and choose an additive or multiplicative model. (Plot your data!)
- Examine the usual diagnostic plots, and plot your residuals as a function of time. Do you need a (different) nonlinear time trend? A transformation of Y?
- Check your residuals for autocorrelation. If it's present, add appropriate lag terms to your model.

The market model

- One final model is of interest: the market model for a stock estimates its volatility/risk relative to the market as a whole
- Important case of regressing one time series on another
- Predict the stock's percentage change (return) as a function of the percentage change of a market index (e.g., DJIA, S&P 500)
- The slope coefficient is the "beta" of the stock:
  - $\beta > 1$ means the stock is riskier/more volatile than the market as a whole
  - $\beta = 1$ means the stock is as risky/as volatile as the market as a whole
  - $\beta < 1$ means the stock is less risky/less volatile than the market as a whole

Market model for Dell stock

First let's calculate the percentage returns:

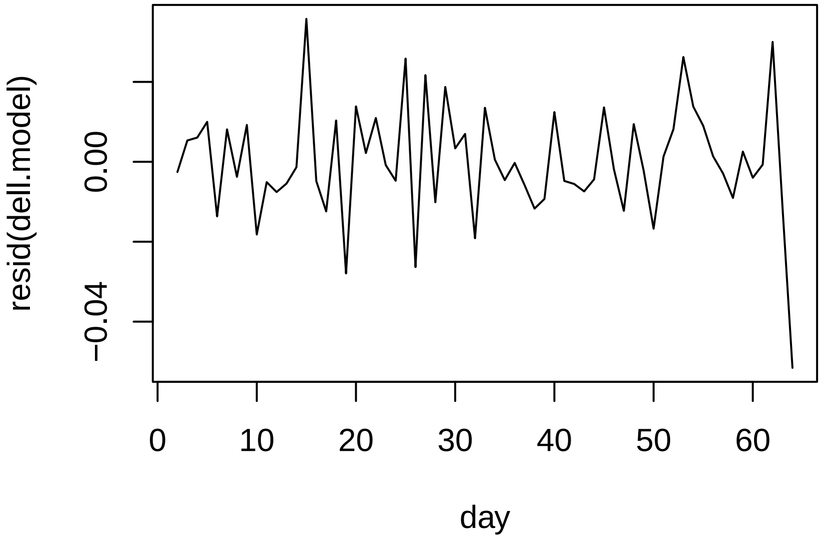
```
dell <- dell %>%
  mutate(dell.return=(dell-lag(dell))/lag(dell),
         SP500.return=(SP500-lag(SP500))/lag(SP500))
```

Now let's predict Dell's returns from the S&P 500:

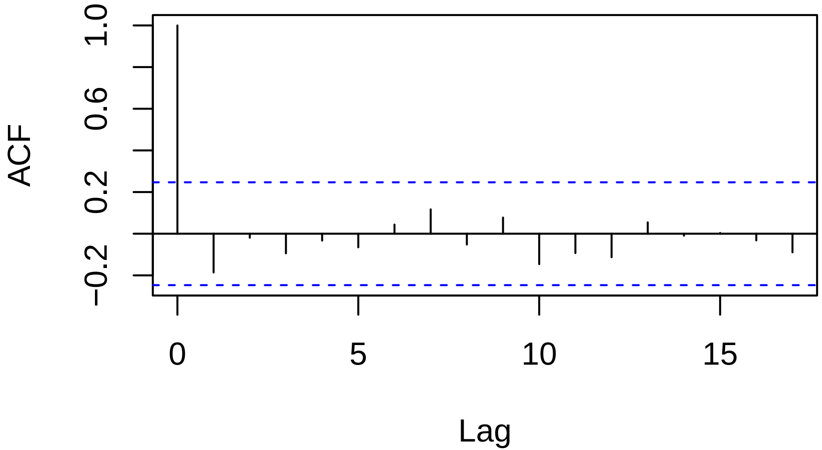
```
dell.model <- lm(dell.return ~ SP500.return, data=dell)
summary(dell.model)


Call:
lm(formula = dell.return ~ SP500.return, data = dell)

Residuals:
     Min       1Q   Median       3Q      Max
-0.05157 -0.00663 -0.00134  0.00912  0.03577

Coefficients:
             Estimate Std. Error t value   Pr(>|t|)
(Intercept)  -0.00217    0.00183   -1.18       0.24
SP500.return  1.09719    0.18946    5.79 0.00000026 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0144 on 61 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.355,Adjusted R-squared:  0.344
F-statistic: 33.5 on 1 and 61 DF,  p-value: 0.000000262
```

Standard diagnostics look pretty good. How about time dependence in residuals?

```
plot(resid(dell.model)~day, data=dell[-1,], type='l')
```



```
acf(resid(dell.model))
```



These look good!

Dell's β

- Dell's β of 1.10 indicates it's about as volatile/risky as the market as a whole
- The R^2 of 0.36 indicates that about 36% of the variability in Dell's stock price can be attributed to the market's movement
- Since $1 - R^2 = 0.64$, about 64% of the variability in Dell's stock price is from Dell-specific (or industry-specific factors)